Arista Access Switch Routing Protocol Flap SOP (KOR-ACC-66)

Node: KOR-ACC-66 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_5800

Description

The failure domain of BGP route churn, OSPF adjacency loss, and ISIS LSP flooding on the Arista Access Switch at Koramangala Datacenter (Node ID: KOR-ACC-66) is a complex phenomenon rooted in the intricate dynamics of routing protocols. The BGP route churn occurs when there is a sudden and excessive change in the routing table, resulting in a cascade of updates that overwhelm the control plane. This, in turn, can lead to OSPF adjacency loss as the switch struggles to maintain stable connections with its neighbors. Meanwhile, the ISIS LSP flooding further exacerbates the situation by creating a deluge of LSPs that the switch is unable to process in a timely manner. This SOP is designed to troubleshoot and resolve this issue by isolating the root cause and implementing corrective measures to stabilize the routing protocols.

Pre-requisite Checks

- check: Verify that the redundant link at Koramangala Datacenter is functioning correctly., status: Required

- check: Ensure that the Arista Access Switch has the latest software image., status: Recommended

- check: Check the temperature and humidity levels at the datacenter to ensure that they are within the recommended operating range., status: Recommended

Potential Root Causes

Cause	Probability
Misconfigured BGP neighbor	High
OSPF neighbor authentication issue	Medium
ISIS LSP flooding due to excessive LSPs	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp neighbors, Expected Output: BGP neighbor is 192.168.1.1, remote AS 65000, local AS 65000, Description: This command displays information about the BGP neighbor, including its IP address, AS number, and local AS number. The output will help identify if there are any misconfigured BGP neighbors.

Command: show ospf neighbors, Expected Output: OSPF neighbor is 192.168.1.1, state FULL, dead time 30, Description: This command displays information about the OSPF neighbor, including its IP address, state, and dead time. The output will help identify if there are any OSPF neighbor authentication issues.
```

```
Command: show isis database, Expected Output: LSP ID: 10000001, LSP Seq: 0x12345678, Description: This command displays information about the ISIS LSP database, including the LSP ID and sequence number. The output will help identify if there are any ISIS LSP flooding issues due to excessive LSPs.
```

Resolution Steps

Step 1: Step: Verify BGP neighbor configuration, Command: show ip bgp neighbors, Description: Verify the BGP neighbor configuration and ensure that it is correct. If there are any issues, update the configuration to resolve the problem.

Step 2: Step: Update OSPF neighbor authentication, Command: config ospf neighbor 192.168.1.1 authentication-key , Description: Update the OSPF neighbor authentication to resolve any issues. Replace with the actual authentication key.

Step 3: Step: Clear ISIS LSP database, Command: clear isis database, Description: Clear the ISIS LSP database to resolve any issues due to excessive LSPs.

Post-Check Verification

```
Command: show ip bgp neighbors, Expected Output: BGP neighbor is 192.168.1.1, remote AS 65000, local AS 65000, Description: Verify that the BGP neighbor configuration is correct.

Command: show ospf neighbors, Expected Output: OSPF neighbor is 192.168.1.1, state FULL, dead time 30, Description: Verify that the OSPF neighbor authentication has been updated correctly.

Command: show isis database, Expected Output: LSP ID: 10000001, LSP Seq: 0x12345678, Description: Verify that the ISIS LSP database has been cleared correctly.
```

Roll-back Procedures

- Step: Restore BGP neighbor configuration, Command: config ip bgp neighbor 192.168.1.1 remote-as 65000, Description: Restore the BGP neighbor configuration to its original state.

- Step: Restore OSPF neighbor authentication, Command: config ospf neighbor 192.168.1.1 authentication-key , Description: Restore the OSPF neighbor authentication to its original state. Replace with the actual authentication key.

- Step: Restore ISIS LSP database, Command: clear isis database, Description: Restore the ISIS LSP database to its original state.